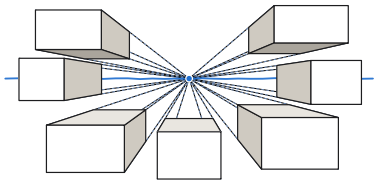
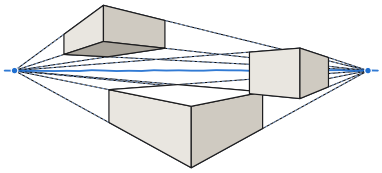
# 7 Types of Perspective

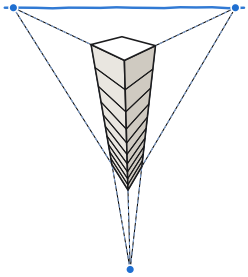
A quick, visual guide to perspective drawing, from 1-point all the way to fisheye. Plus the 5 core elements, a quick-pick map and a 7-day challenge.



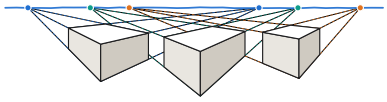
**1** 1-Point



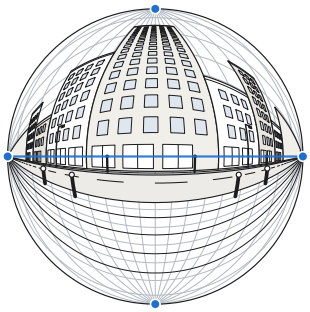
**2** 2-Point



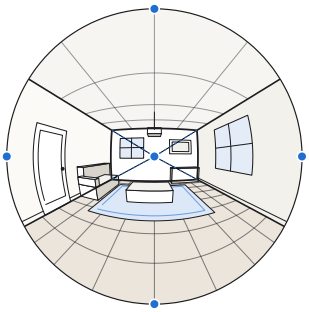
**3** 3-Point



**4** Multi-Point



**5** 4-Point



**6** 5-Point (Fisheye)

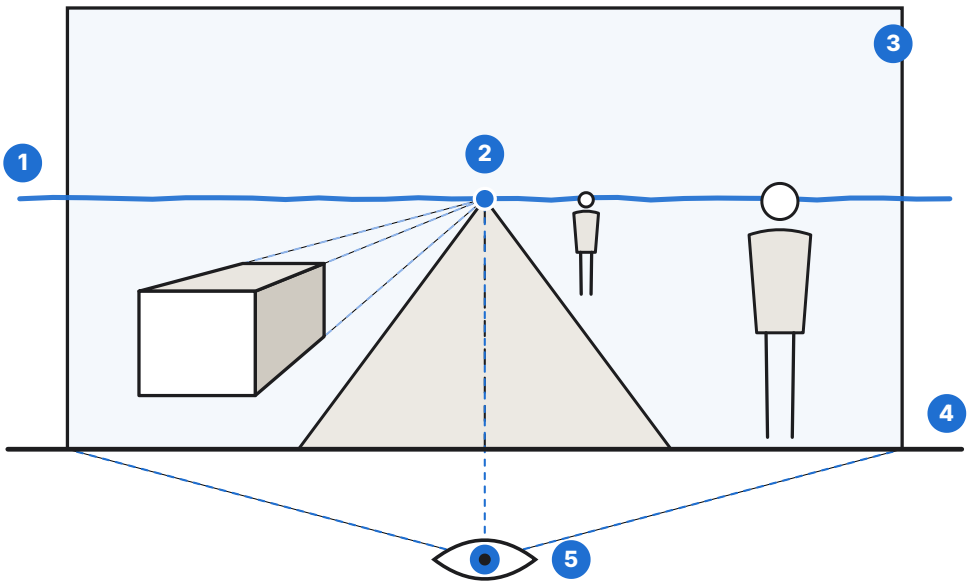


**7** Aerial (Atmospheric) Perspective

# Before You Start: The 5 Core Elements

The building blocks of every perspective drawing

Every perspective drawing, however complex, is built on these five elements. Learn them once, and every type in this guide gets much easier to understand.




**1 Horizon Line** Eye Level
An imaginary horizontal line at the exact height of your eyes. Anything above it, you see from below. Anything below it, you see from above.

**2 Vanishing Point** VP
A point on the horizon where parallel lines going into the distance meet. Every set of parallel lines has its own vanishing point.

**3 Picture Plane**
An imaginary sheet of glass between you and the scene. Your paper is that glass: the scene gets traced onto it.

**4 Ground Line**
The bottom edge of the picture plane, where the glass touches the ground. Your measurements start here.

**5 Station Point** Viewer
Where the viewer's eye is. Step closer or farther and the spacing of the vanishing points, and the amount of distortion, changes.

**The golden rule**

If you draw while standing, the eyes of every person of similar height standing on the same ground, near or far, sit right on the horizon line. Only the size of their bodies changes.

# The Perspective Map

Which type for which scene?

### Linear
1, 2, 3 & multi-point

Straight lines, with vanishing points on the horizon. Includes 1-, 2-, 3-point and multi-point.

### Curvilinear
4 & 5-point

Curved lines and a very wide view. Includes 4- and 5-point (even 6-point).

### Atmospheric
Aerial perspective

Depth through value, color and contrast, with no vanishing points at all.

| Type | Vanishing points | Best for | Page |
|---|---|---|---|
| **1-Point** | 1 | Streets, roads, hallways, rooms seen head-on, railway tracks | 4 |
| **2-Point** | 2 | Building corners, exteriors, furniture and objects at an angle | 5 |
| **3-Point** | 3 | Skyscrapers, views from above or below, dramatic scenes | 6 |
| **Multi-Point** | 4+ | Objects placed at different angles, old towns, irregular streets | 7 |
| **4-Point** | 4 (curved) | Panoramas, very wide or very tall spaces | 8 |
| **5-Point** | 5 (curved) | Fisheye effect, showing an entire room at once | 9 |
| **Aerial** | None | Landscapes and mountains; depth through value, color and contrast | 10 |

## Quick pick

Facing the scene straight on? → **1-Point**

Looking at the corner of an object? → **2-Point**

Looking from very high or very low? → **3-Point**

Objects turned at different angles? → **Multi-Point**

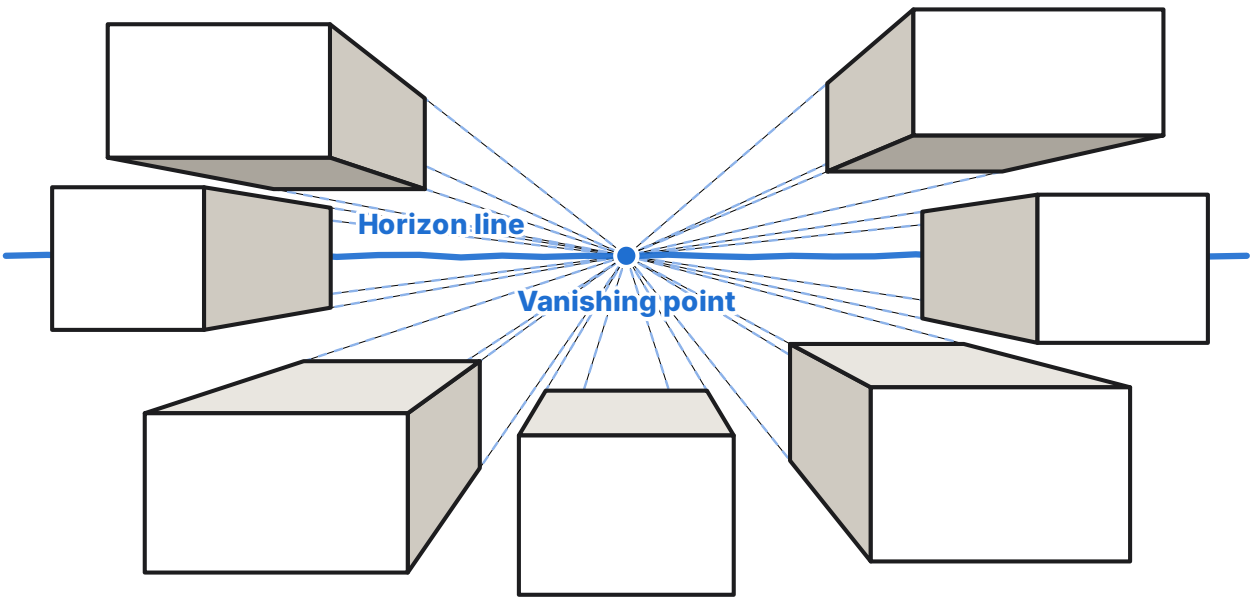
Scene too wide to fit in the frame? → **4- or 5-Point**

A landscape with lots of depth? → **Add aerial to any type**

# 1 1-Point Perspective

The straight-on view

Vanishing points: 1




## What is it?

When one face of an object is facing you directly, every line going into depth meets at a single point on the horizon. Horizontal lines stay horizontal, and vertical lines stay vertical.

## When to use it

- Streets & roads
- Hallways
- Rooms seen head-on
- Railway tracks
- Shelves & bookcases

## Step by step

1. Draw the horizon line and place one vanishing point on it.
2. Draw the front face of the object as a flat rectangle or square.
3. Connect the corners of that face to the vanishing point with light guide lines.
4. Close the depth with one horizontal and one vertical line, then erase the extra guides.

**Quick exercise:** Put a vanishing point in the middle of your page and draw 9 boxes: three above the horizon, three on it and three below it. Notice which faces you can see on each one.

**Common mistake**

Too much depth. A box pulled too far toward the vanishing point looks stretched and unreal.

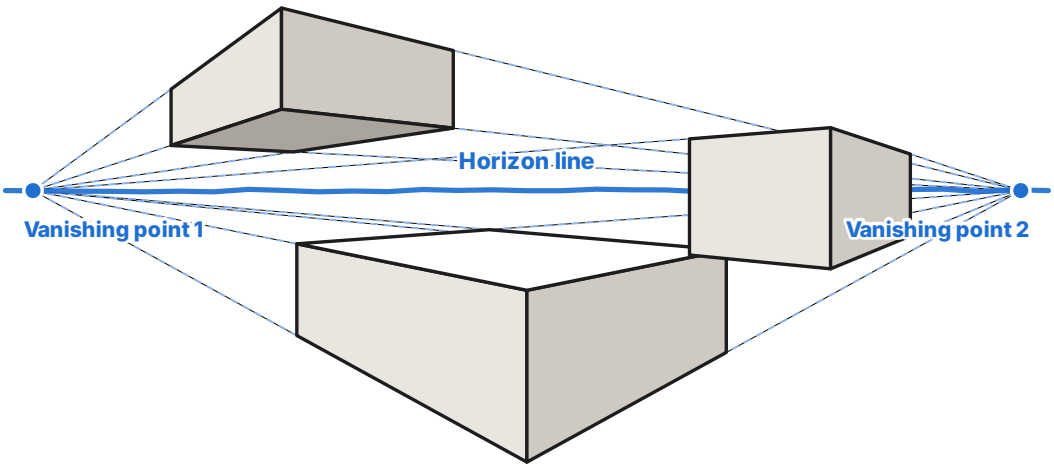
**Pro tip**

Don't put the vanishing point dead center. Shifting it a little makes the composition feel more natural and interesting.

# 2 2-Point Perspective

The corner view

Vanishing points: 2




## What is it?

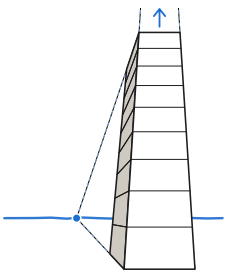
When you look at the corner of an object, you get two sets of depth lines: one runs to the left vanishing point, the other to the right. Only vertical lines stay vertical.

## When to use it

Building exteriors · Street corners · Furniture · Product design

## Step by step

1. Draw the horizon and place two vanishing points as far apart as you can.
2. Draw the nearest vertical edge of the object.
3. Connect the top and bottom of that edge to both vanishing points.
4. Add two more vertical edges to set the width, then connect the new corners back to the vanishing points.



### Special case: Vertical 2-Point

One vanishing point on the horizon for depth, and one above the frame for the vertical lines. Stand in front of a tall tower and tilt your head up: the vertical edges lean toward each other as they rise.

### Common mistake

Vanishing points too close together. You get sharp, distorted corners. More distance means a more natural image.

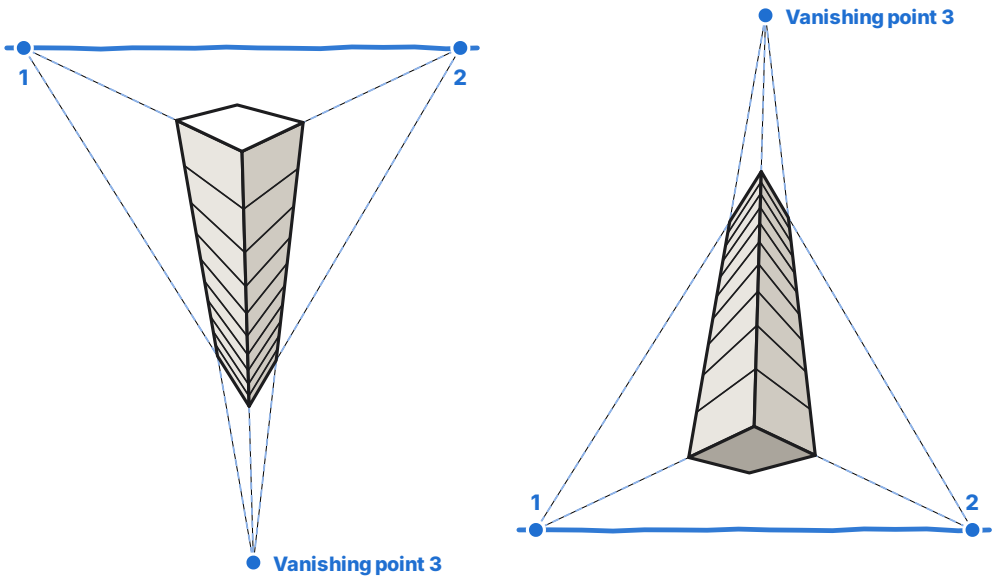
### Pro tip

If your vanishing points don't fit on the paper, tape the paper to your desk and mark the points on the desk itself.

# 3 3-Point Perspective

The dramatic view

Vanishing points: 3




Bird's-eye view (from above) Worm's-eye view (from below)

## What is it?

On top of the two vanishing points on the horizon, there's a third one above or below the picture where the vertical lines meet. The result is a strong sense of height and a dramatic viewpoint.

## When to use it

Skyscrapers | City from above | Looking up at a tower | Posters & cinematic shots

## Step by step

1. Place two vanishing points on the horizon.
2. Put the third vanishing point far from the horizon: below it for a view from above, above it for a view from below.
3. Find the nearest corner of the object and connect it to all three points.
4. Draw the vertical edges leaning toward the third point instead of straight up and down.

**Quick exercise:** Draw a simple cube three times, moving the third point farther away each time. Which distance looks the most natural?

**Common mistake**

Third point too close. The building turns into a sharp cone. The three points should form a large triangle.

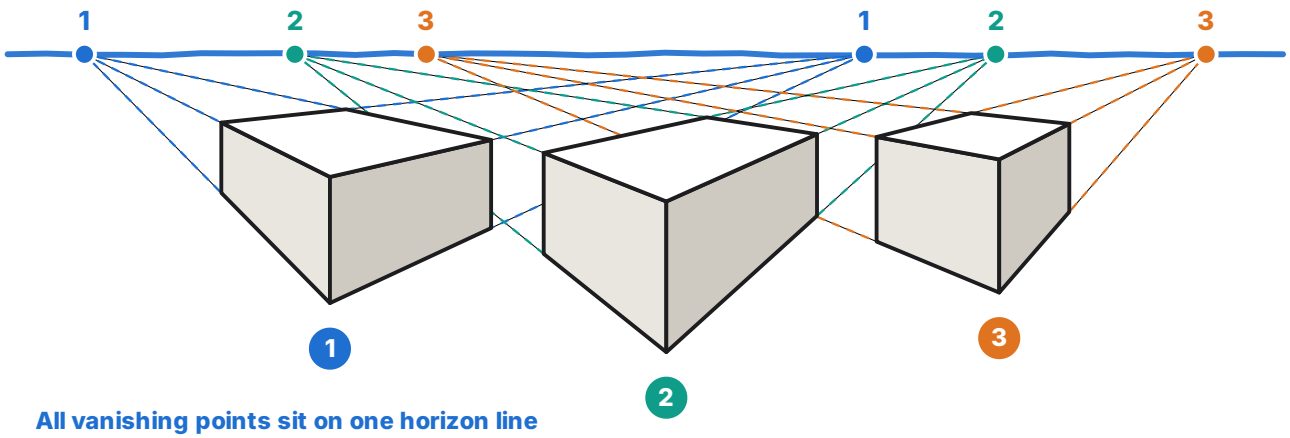
**Pro tip**

For a bird's-eye view, push the horizon to the top of the page; for a worm's-eye view, to the bottom. That one decision changes the whole mood.

# 4 Multi-Point Perspective

The real-world view

Vanishing points: 4+




All vanishing points sit on one horizon line

## What is it?

In the real world, not everything is lined up. Every object turned at a different angle gets its own pair of vanishing points, but all of them sit on the same horizon line.

## When to use it

- Furniture around a room
- Houses in a neighborhood
- Rotated volumes in architecture
- Irregular cityscapes

## Step by step

1. Draw one shared horizon line for the whole scene.
2. Draw the first object in regular 2-point perspective.
3. For every object at a different angle, add a new pair of vanishing points on the same horizon.
4. Notice: the closer one point of a pair is to the center, the farther away its partner goes.

**Quick exercise:** Place three books or boxes on a table at different angles and draw them from life. Find the pair of vanishing points for each one.

**Common mistake**

Putting vanishing points on different horizon lines. On flat ground, all horizontal vanishing points share one horizon.

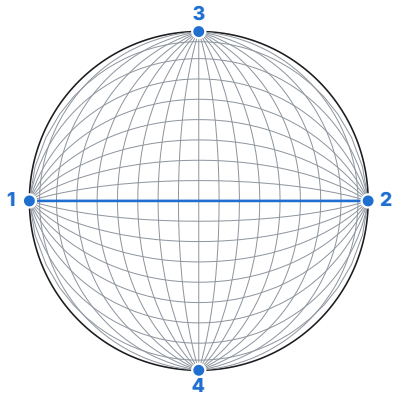
**Pro tip**

When you're starting out, mark each object's vanishing points in a different color so they don't get lost among the lines.

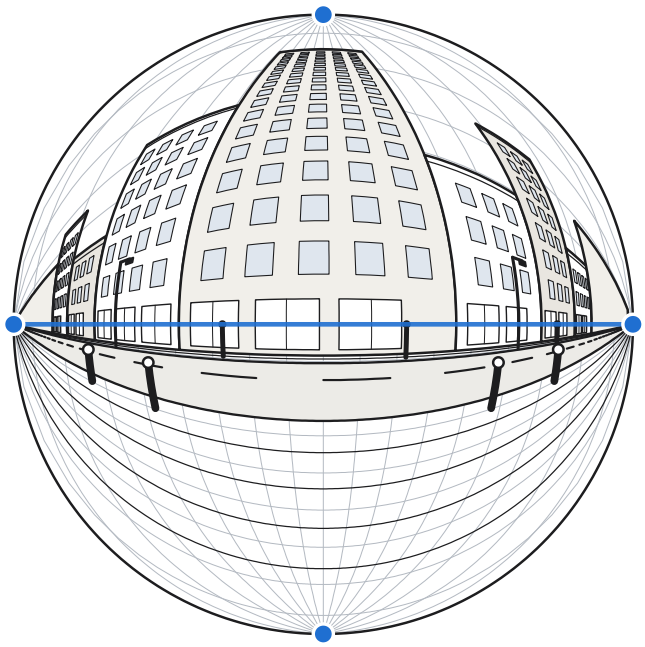
# 5 4-Point Perspective

The panoramic view · Curvilinear

Vanishing points: 4 (curved)




The grid and its 4 vanishing points



A street drawn on the same grid

## What is it?

This is curvilinear perspective: straight lines appear curved, as if you squeezed a very wide scene into one frame. Two vanishing points sit on the left and right, two on the top and bottom, all on the edge of a circle.

## When to use it

Panoramas · Very wide or tall spaces · Illustration & comics · Fantasy scenes

## Step by step

1. Draw a circle. Its horizontal diameter is your horizon line.
2. Place four vanishing points on the left, right, top and bottom of the circle.
3. Build a curved grid: arcs through the left and right points, and arcs through the top and bottom points.
4. Draw your scene on the grid. Only the lines that pass through the center stay straight.

**Quick exercise:** Draw just the grid: a circle, four points and five arcs in each direction. Once it's clean, design a row of buildings on it.

**Common mistake**

Reaching for a ruler. Almost no line is straight in this perspective, so freehand and a guide grid are your best tools.

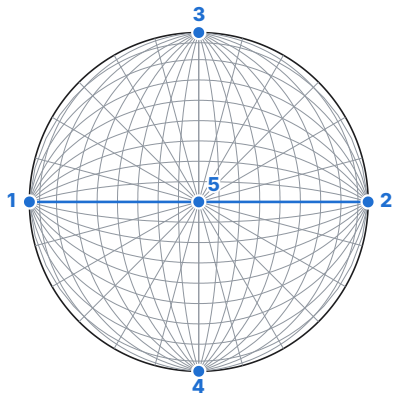
**Pro tip**

Draw the grid carefully once, make a copy and slip it under thin paper for every practice session after that.

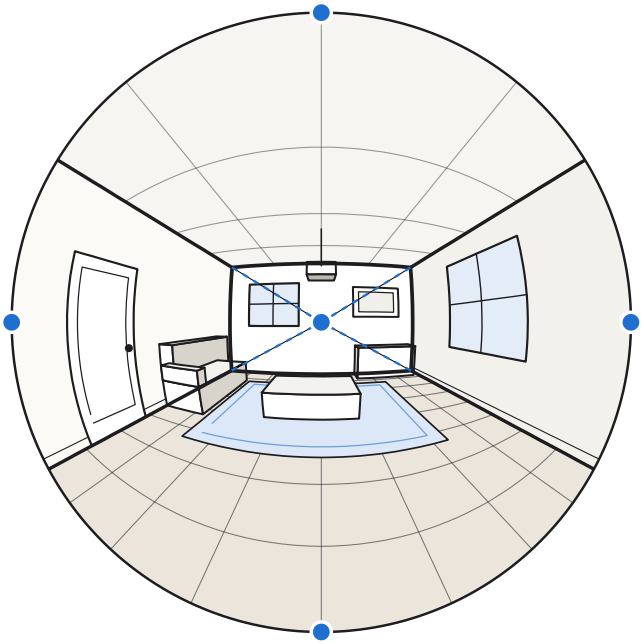
# 6 5-Point Perspective

The fisheye view

Vanishing points: 5 (curved)




The grid and its 5 vanishing points



A room drawn on the same grid

## What is it?

It's 4-point perspective plus one more vanishing point in the center of the circle. This is the "fisheye" effect, showing almost 180° of a scene at once.

## When to use it

A whole room from the middle | Action scenes | Extreme depth | Creative illustration

## Step by step

1. Draw the circle and the four edge points, just like in 4-point.
2. Place the fifth point in the center. All depth lines run straight into it, like spokes.
3. Add the curved grid for horizontal and vertical lines.
4. Draw the facing wall near the center, small and almost flat. The closer you get to the edge, the stronger the curve.

**Quick exercise:** Stand in the middle of your room. Draw the wall in front of you at the center of the circle, then add the floor, ceiling and side walls around it.

**Common mistake**

Curving every line. Lines that run to the center stay straight; only the others bend, more and more toward the edge.

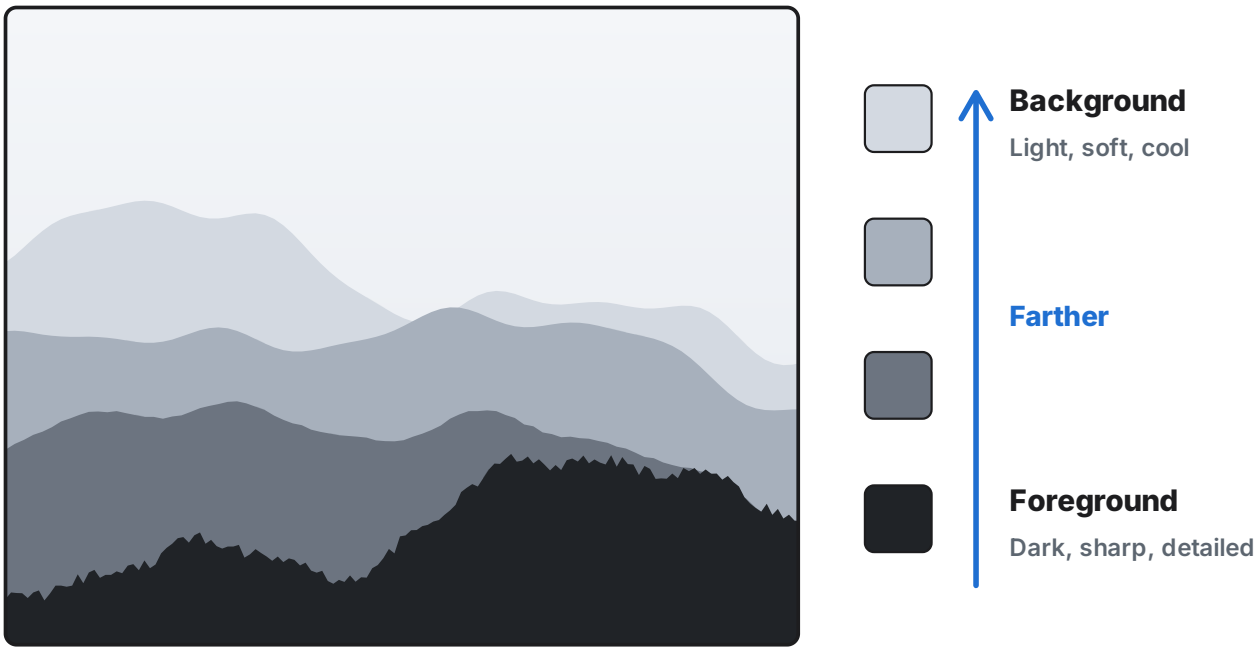
**Pro tip**

Trace the main lines of a fisheye photo with tracing paper. It's the fastest way to understand the logic of the grid.

# 7 Aerial Perspective

Depth through air · Atmospheric

No vanishing points




## What is it?

This one doesn't use lines or vanishing points. It uses air! Between your eye and distant objects there's a layer of air, moisture and dust that makes them look lighter, softer and cooler.

## When to use it

- Landscapes & mountains
- Cities in fog
- Outdoor scenes
- Depth without lines

## Step by step

1. Split the scene into three layers: foreground, middle ground and background.
2. Start with the background, using the lightest value and the least detail.
3. Move forward layer by layer, increasing value and contrast.
4. Save the darkest values and sharpest details for the foreground only.

**Quick exercise:** With one pencil, make a 4-step value scale from light to dark, then use those same four values for four layers of mountains.

**Common mistake**

Too much detail in the background. Drawing every window or leaf in the distance kills the depth.

**Pro tip**

Squint your eyes. Value differences become much easier to see. Combine this with any other type of perspective.

# The 7-Day Perspective Challenge

One type a day, just 20 minutes. Tick off each day when you're done.

| | | | |
|---|---|---|---|
| **Day 1** | **1-Point** | Draw a hallway or a street seen head-on. | ☐ |
| **Day 2** | **2-Point** | Draw 10 boxes: above, on and below the horizon. | ☐ |
| **Day 3** | **3-Point** | Draw a tall tower from a bird's-eye view. | ☐ |
| **Day 4** | **Multi-Point** | Arrange a few boxes or houses at different angles on one ground. | ☐ |
| **Day 5** | **4-Point** | Draw the curved grid and build a street or wall on it. | ☐ |
| **Day 6** | **5-Point** | Draw your own room with a fisheye effect. | ☐ |
| **Day 7** | **Aerial** | Draw a mountain landscape with 4 value layers. | ☐ |

## Before every drawing, ask yourself

- ☐ Where are my eyes? (Draw the horizon line first.)
- ☐ Where is the nearest edge or corner of the object?
- ☐ How many sets of parallel depth lines do I have? (That's the number of vanishing points.)
- ☐ Are my vanishing points far enough apart?

## Show us your drawings!

Snap a photo of your challenge drawings and tag us in your story or post. New drawing and perspective lessons go up on the page every day.

**@perspective.sketch**

**Coming soon: the complete step-by-step perspective course**